

# Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

---

Gajavada Sanjeevkumar

September 9, 2026

# Problem Statement

## Problem

Draw a hypocycloid of a circle of diameter 50 mm, which rolls inside a circular arc of radius 85 mm for one revolution. Also, draw a tangent and a normal to the epicycloid at a point 50 mm from the centre of the directing circle.

## Calculated Data

$$R = 85 \text{ mm},$$

$$OC_0 = R - r = 85 - 25 = 60 \text{ mm},$$

$$\angle AOB = 360^\circ \frac{r}{R} = 360^\circ \frac{25}{85} = 105.882353^\circ,$$

$$\Delta\theta = \frac{105.882353^\circ}{12} = 8.823529^\circ,$$

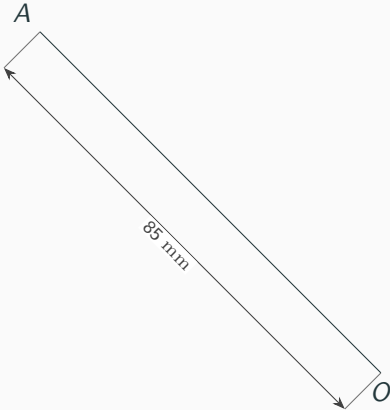
$$r = \frac{50}{2} = 25 \text{ mm},$$

$$AC_0 = 25 \text{ mm},$$

Generating-circle division =  $30^\circ$ .

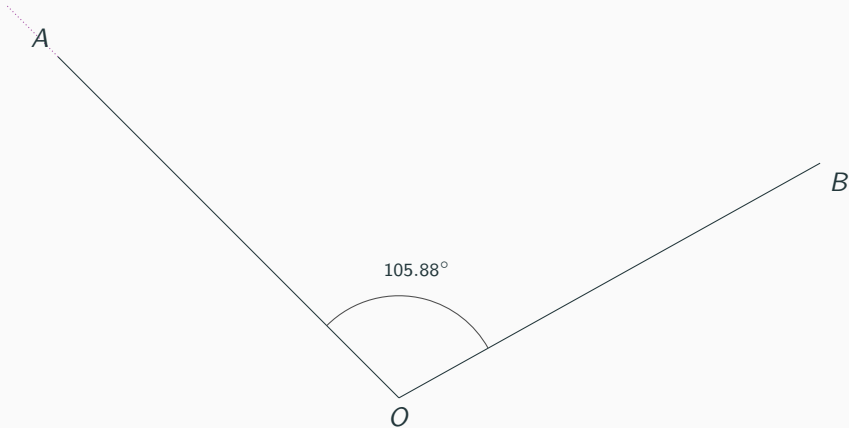
## Step 1: Draw $OA$

$OA = 85 \text{ mm.}$



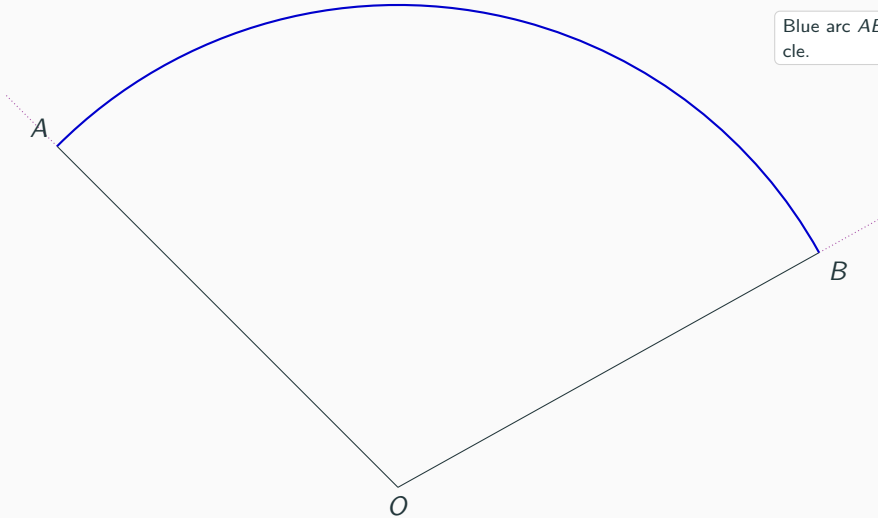
## Step 2: Draw $OB$ at $105.88^\circ$

$$\angle AOB = 105.88^\circ.$$

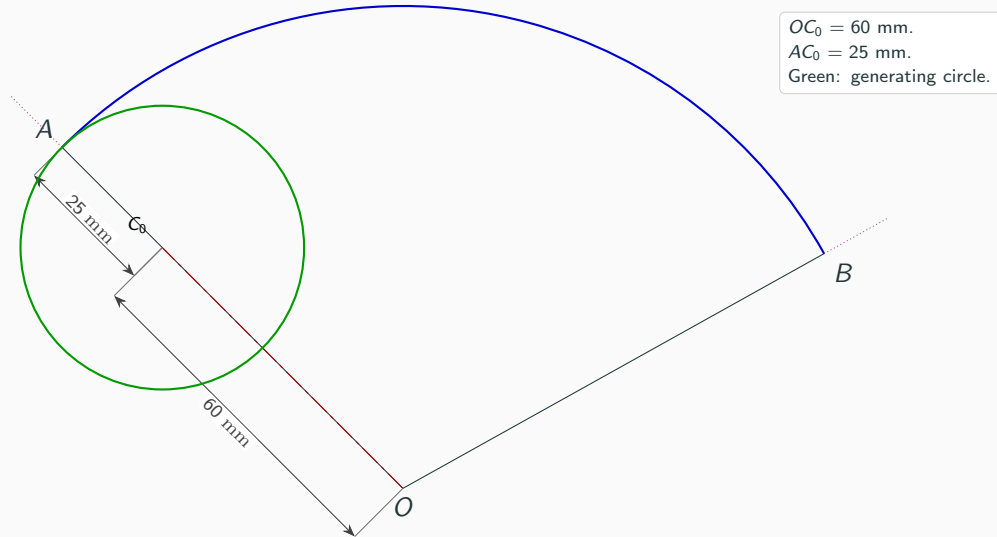


### Step 3: Draw the directing arc $AB$

Blue arc  $AB$ : directing circle.

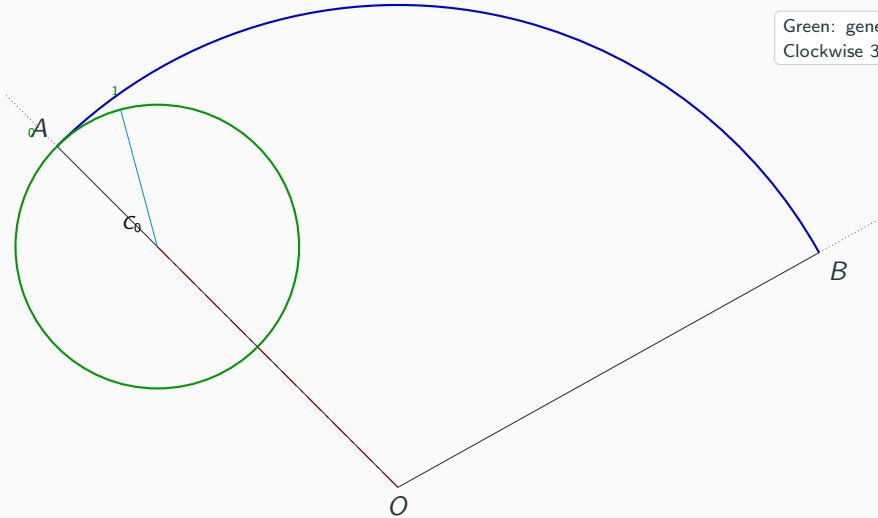


## Step 4: Locate $C_0$ and the generating circle



## Step 5: Divide the generating circle into 12 equal parts

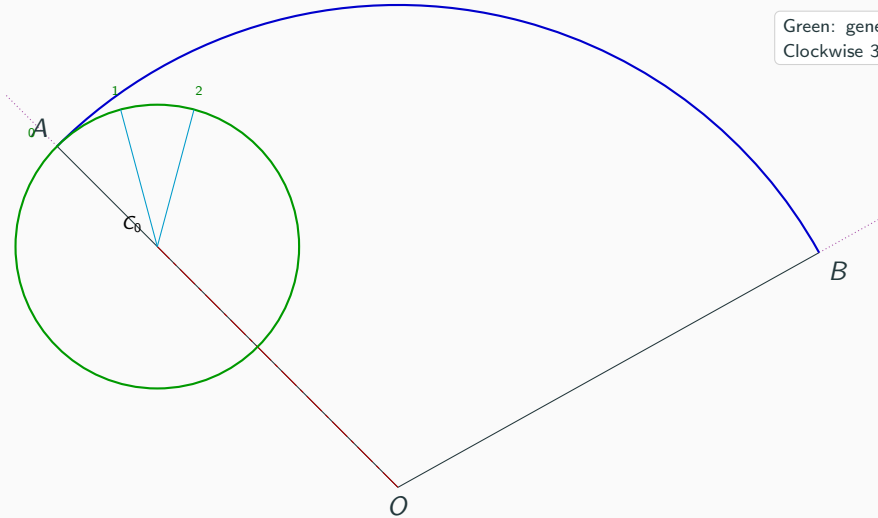
Green: generating circle.  
Clockwise  $30^\circ$  divisions.



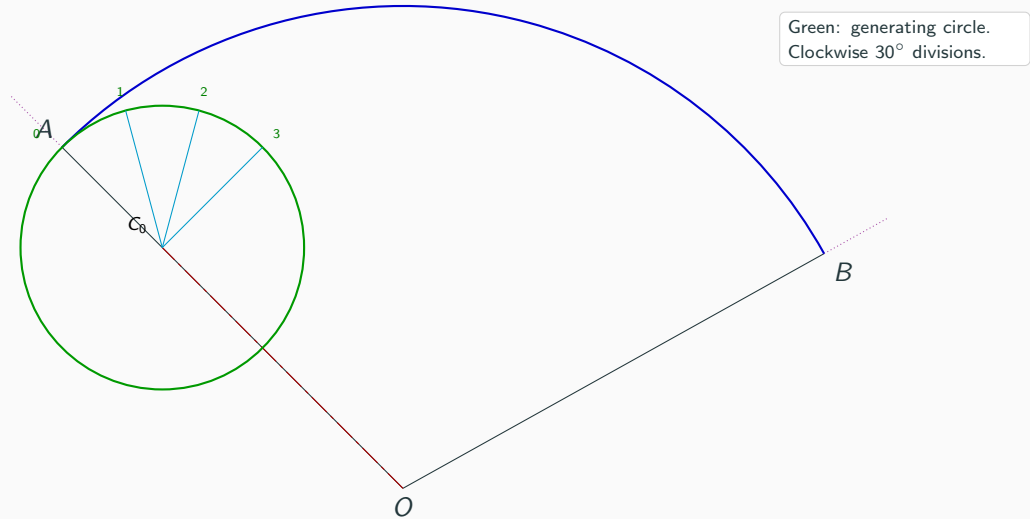


## Step 5: Divide the generating circle into 12 equal parts

Green: generating circle.  
Clockwise  $30^\circ$  divisions.

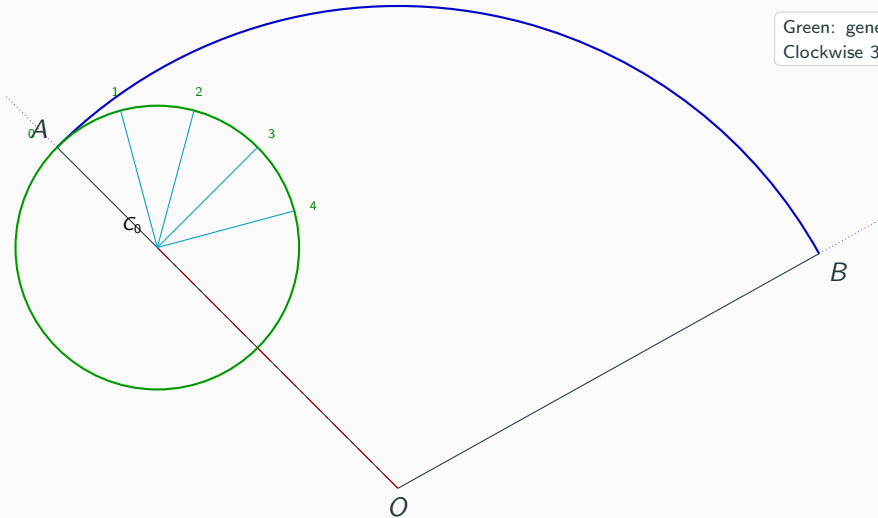


## Step 5: Divide the generating circle into 12 equal parts

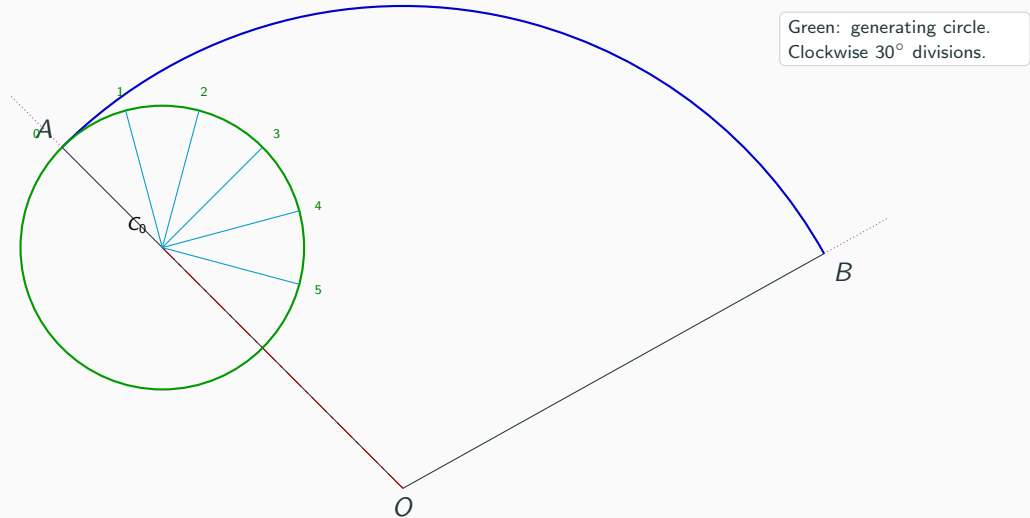


## Step 5: Divide the generating circle into 12 equal parts

Green: generating circle.  
Clockwise  $30^\circ$  divisions.

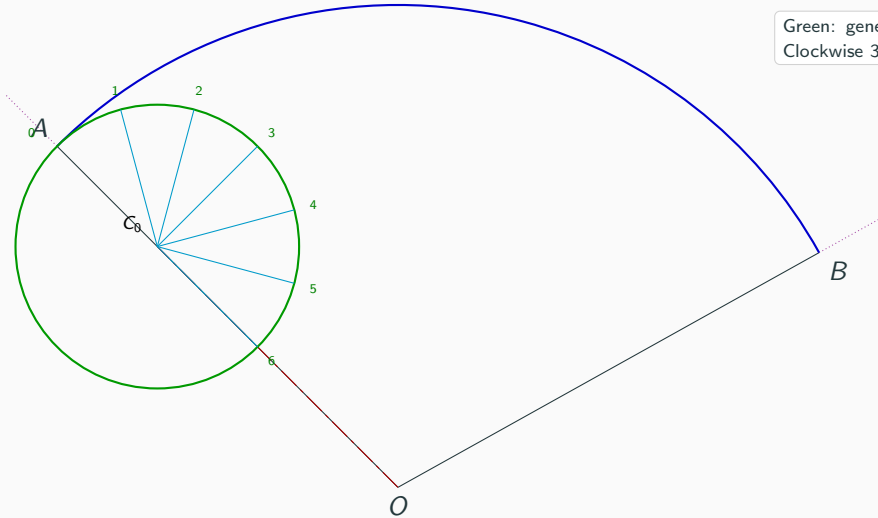


## Step 5: Divide the generating circle into 12 equal parts

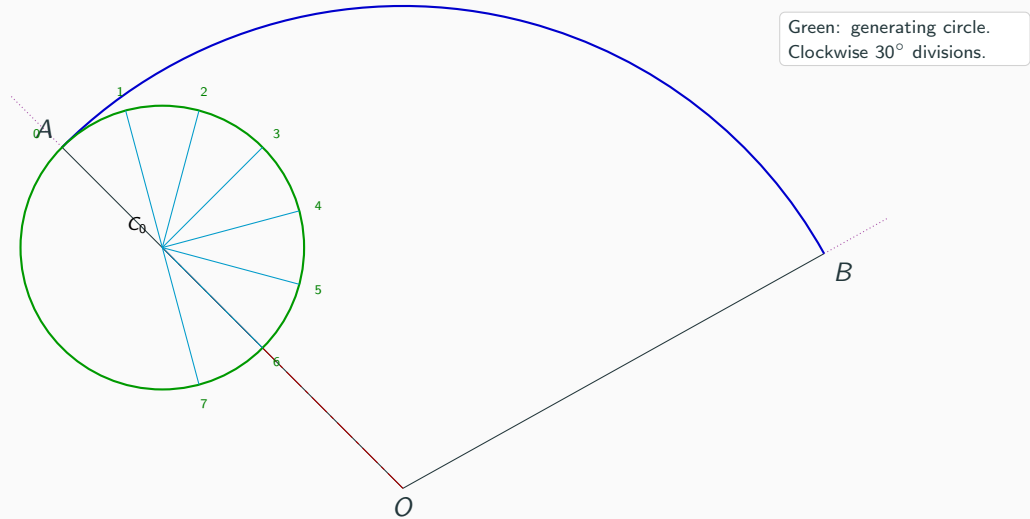


## Step 5: Divide the generating circle into 12 equal parts

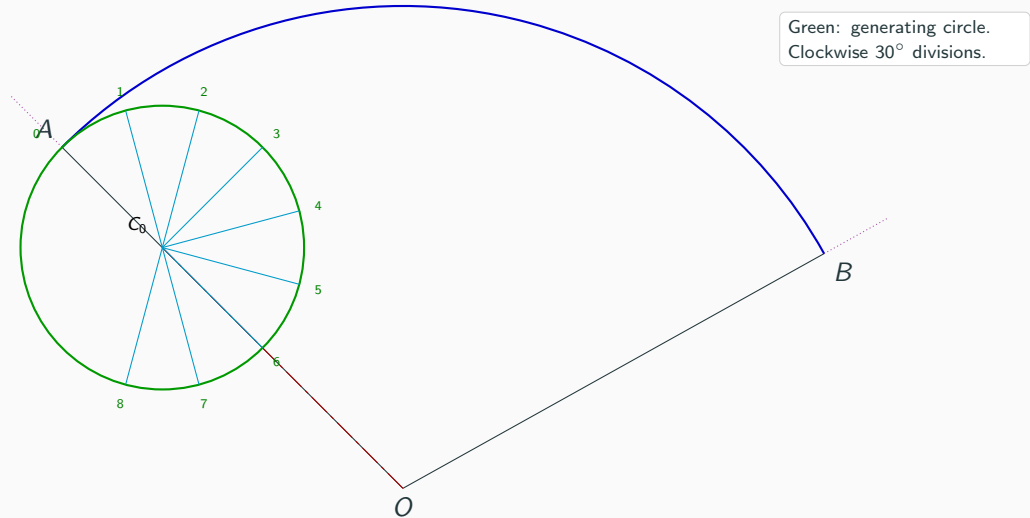
Green: generating circle.  
Clockwise  $30^\circ$  divisions.



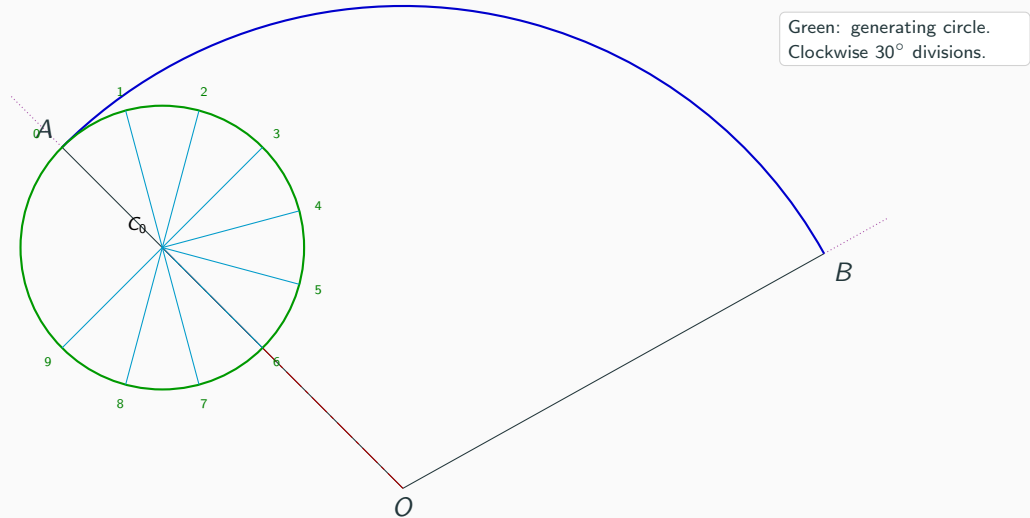
## Step 5: Divide the generating circle into 12 equal parts



## Step 5: Divide the generating circle into 12 equal parts

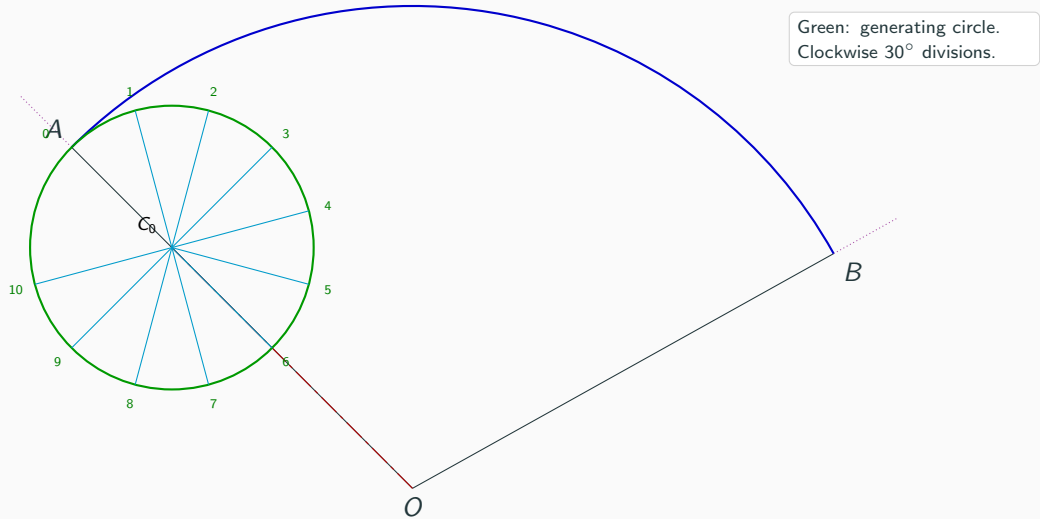


## Step 5: Divide the generating circle into 12 equal parts

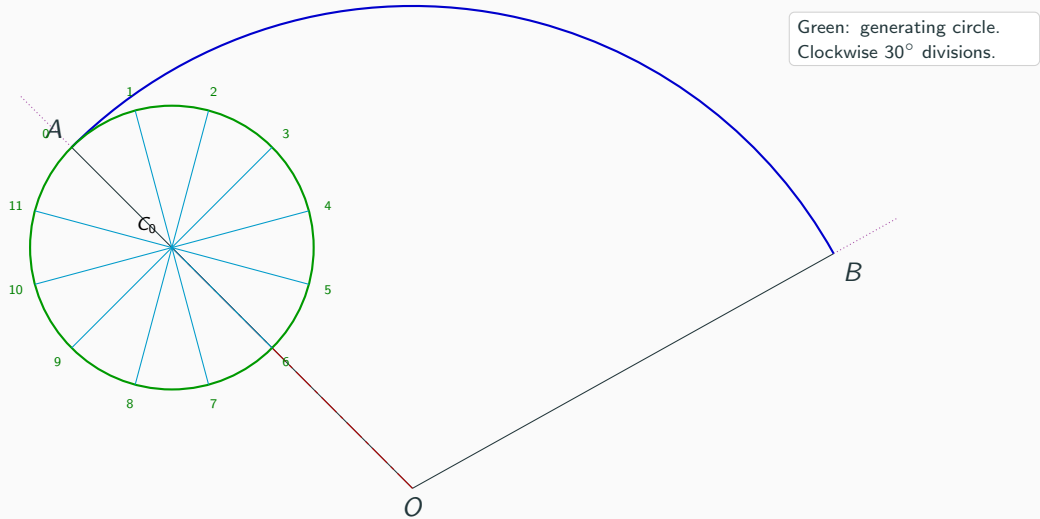




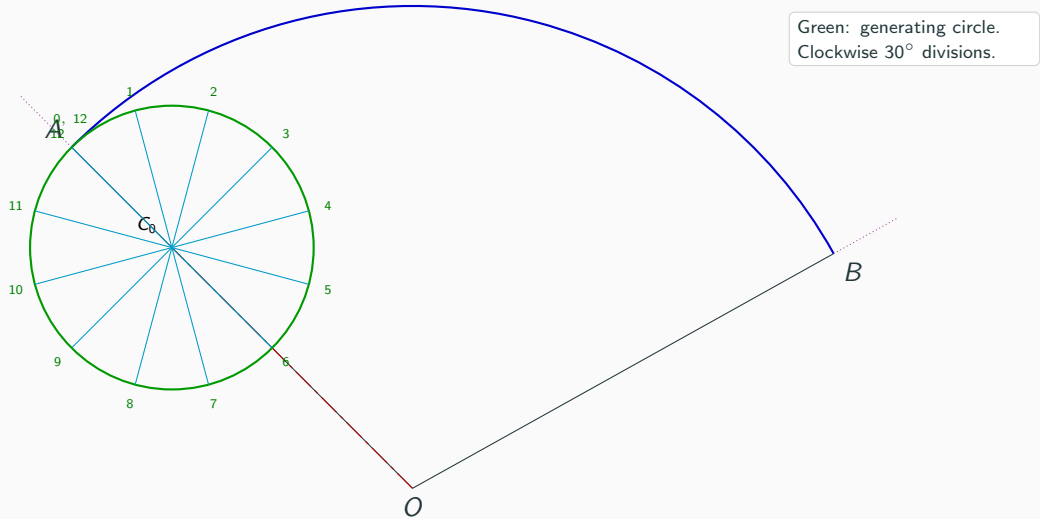
## Step 5: Divide the generating circle into 12 equal parts



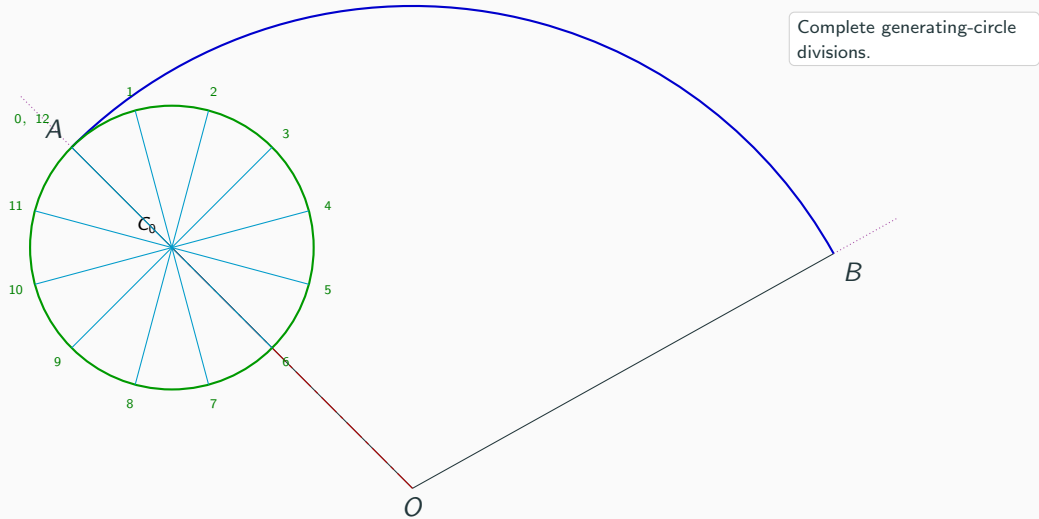
## Step 5: Divide the generating circle into 12 equal parts



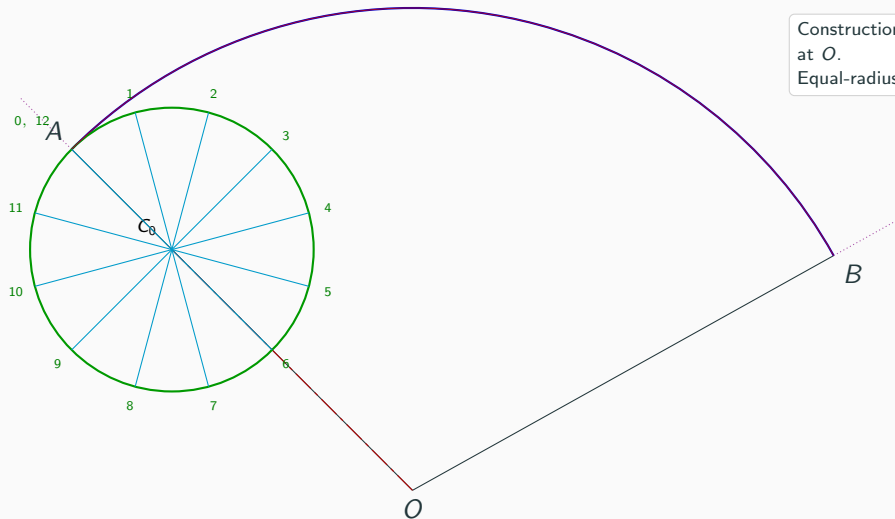
## Step 5: Divide the generating circle into 12 equal parts



## Step 6: Complete the generating-circle divisions

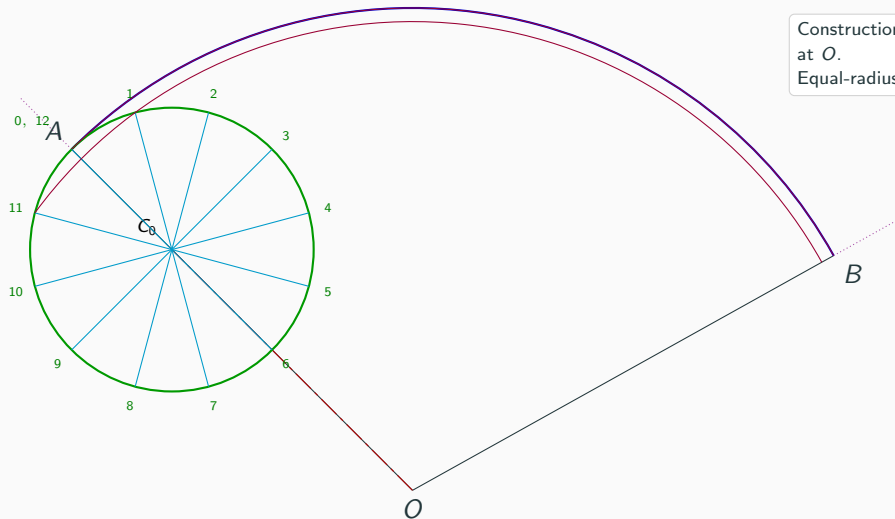


## Step 7: Draw concentric construction arcs



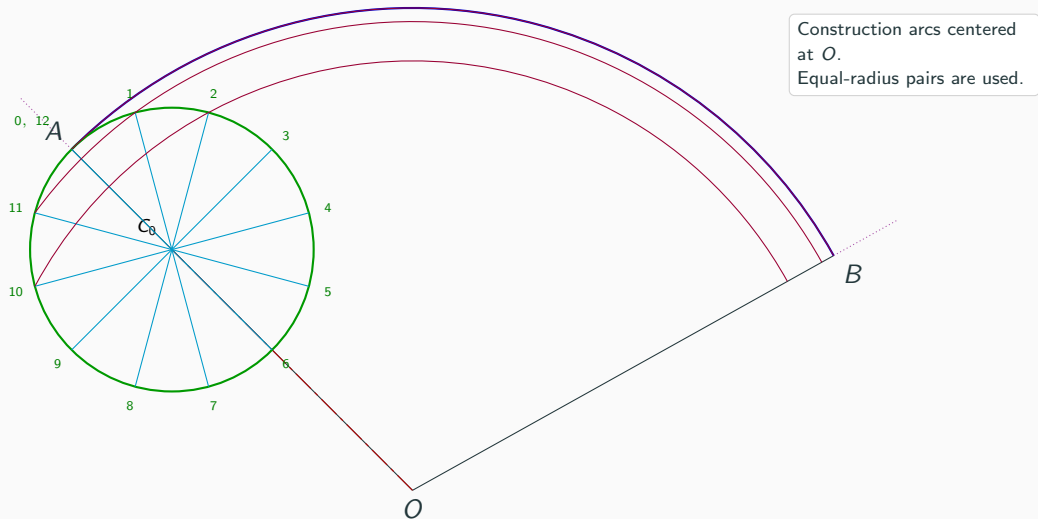
Construction arcs centered at  $O$ .  
Equal-radius pairs are used.

## Step 7: Draw concentric construction arcs

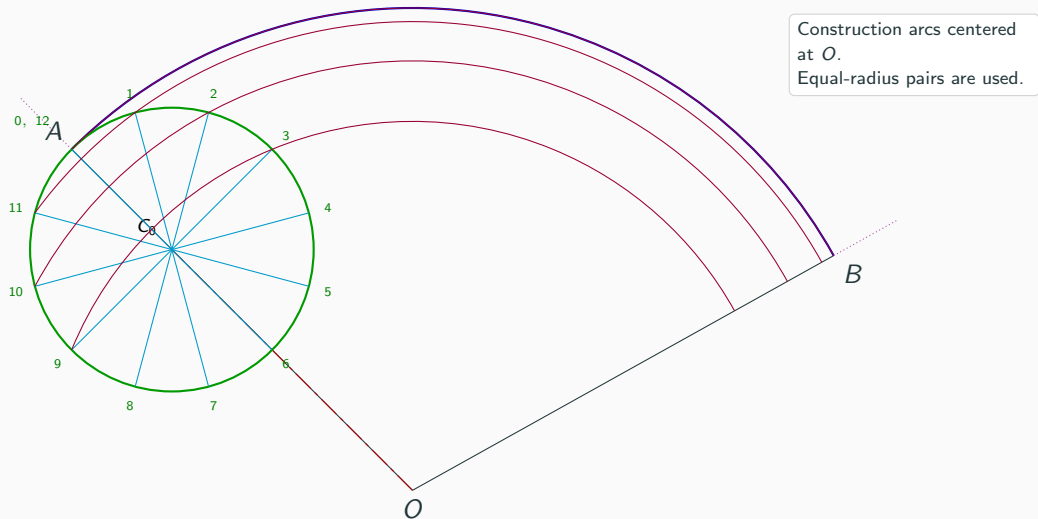


Construction arcs centered at  $O$ .  
Equal-radius pairs are used.

## Step 7: Draw concentric construction arcs

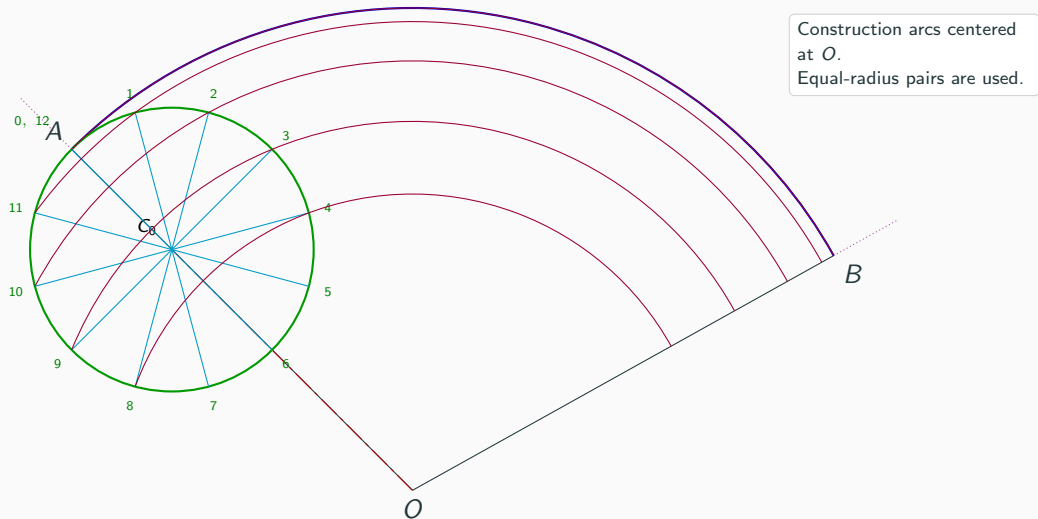


## Step 7: Draw concentric construction arcs

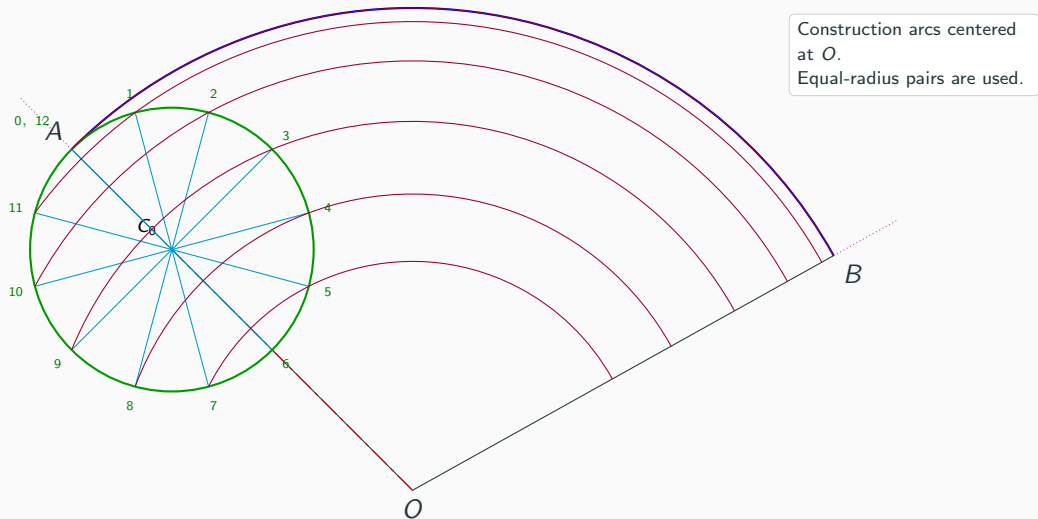




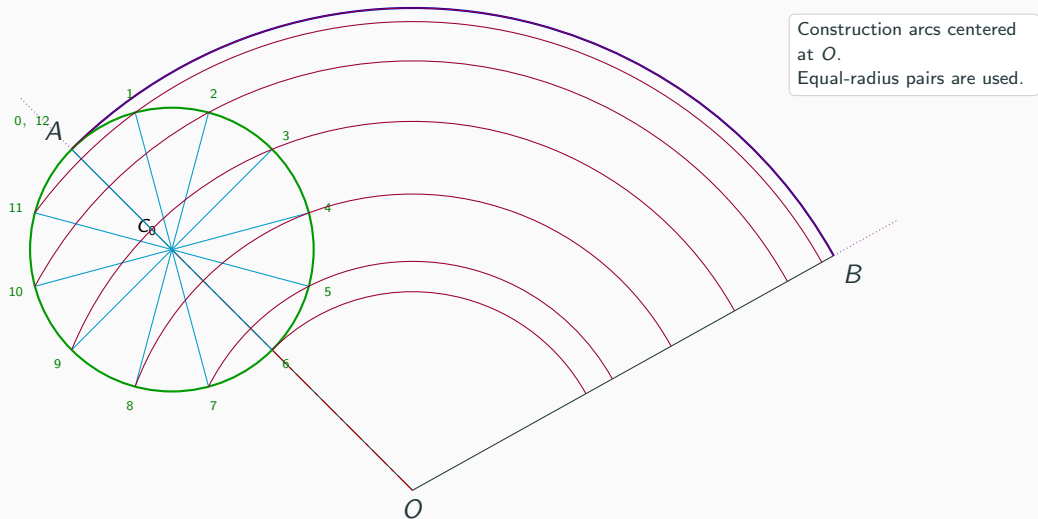
## Step 7: Draw concentric construction arcs



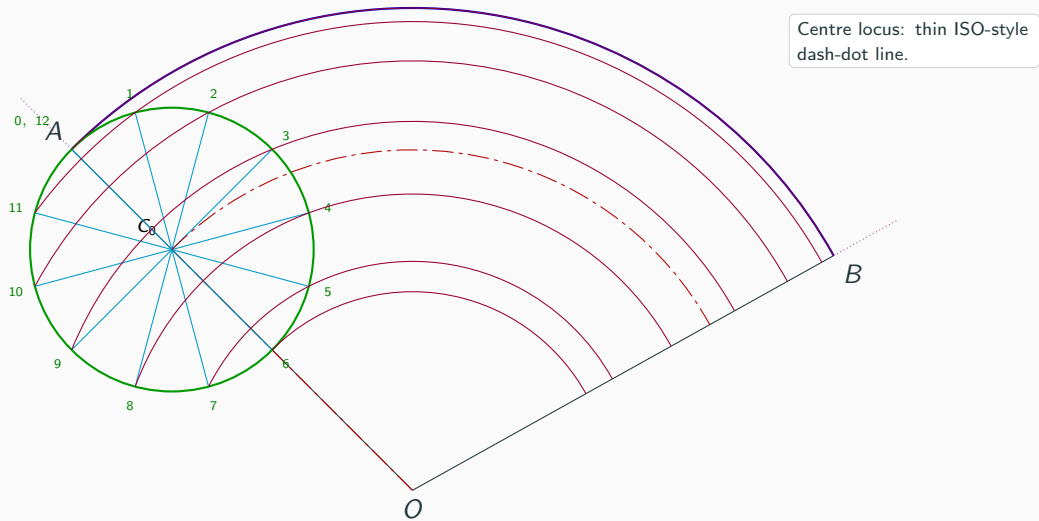
## Step 7: Draw concentric construction arcs



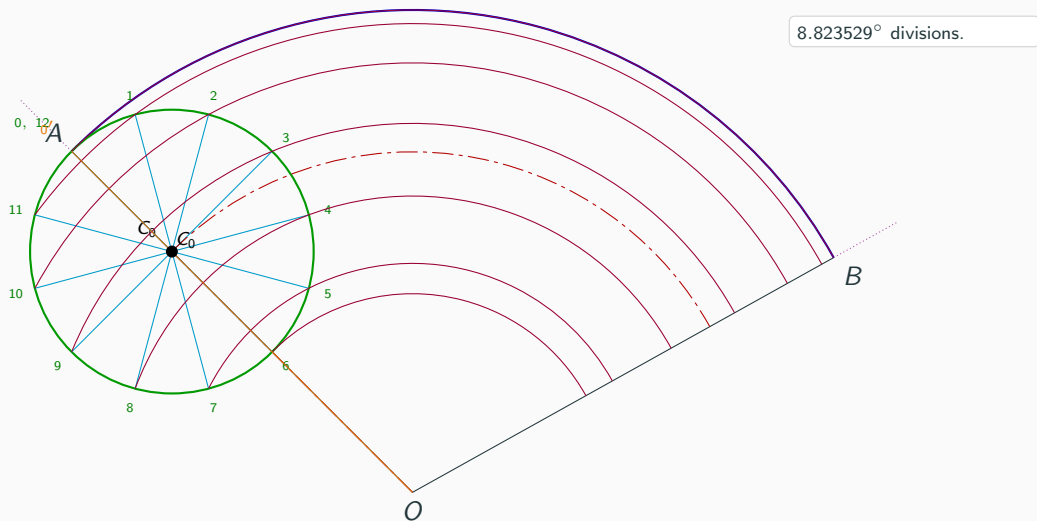
## Step 7: Draw concentric construction arcs



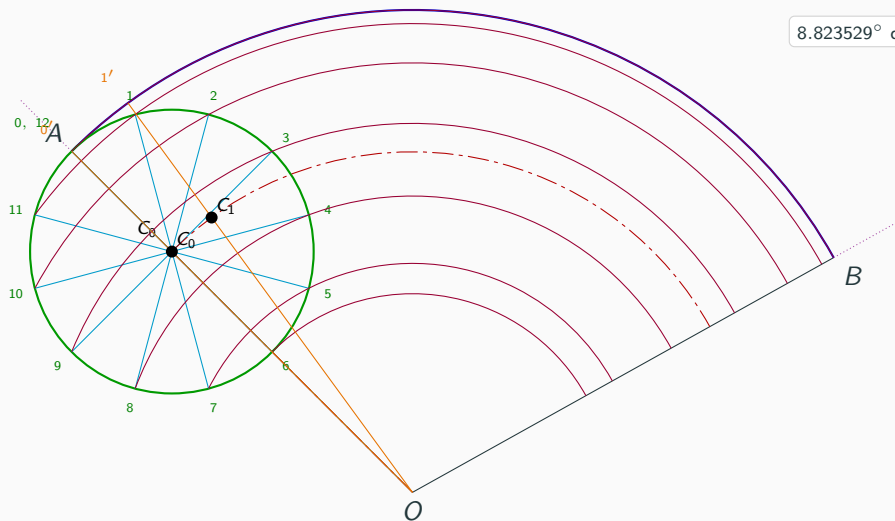
## Step 8: Draw the centre arc through $C_0$



## Step 9: Divide $\angle AOB$ into 12 equal parts

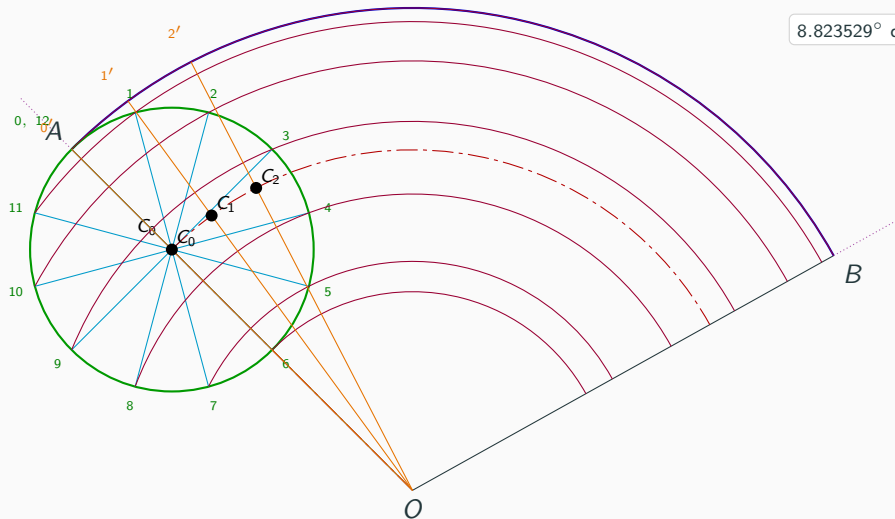


## Step 9: Divide $\angle AOB$ into 12 equal parts

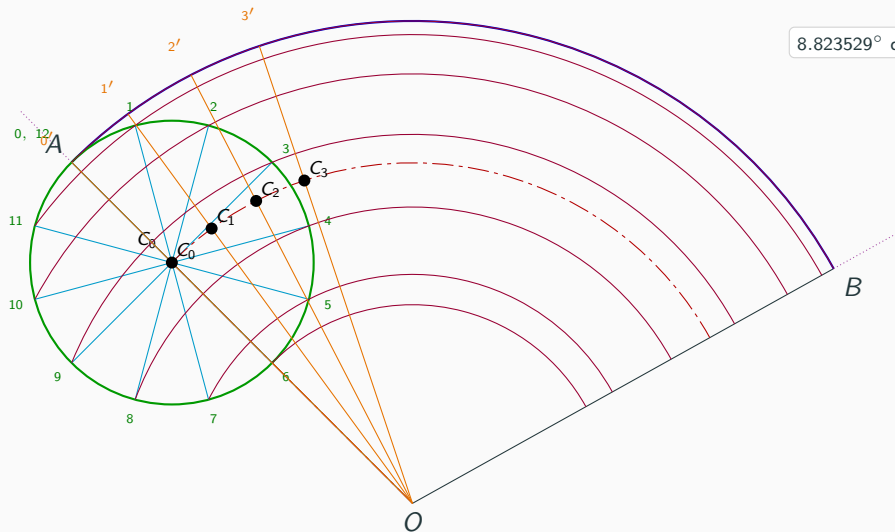


8.823529° divisions.

## Step 9: Divide $\angle AOB$ into 12 equal parts



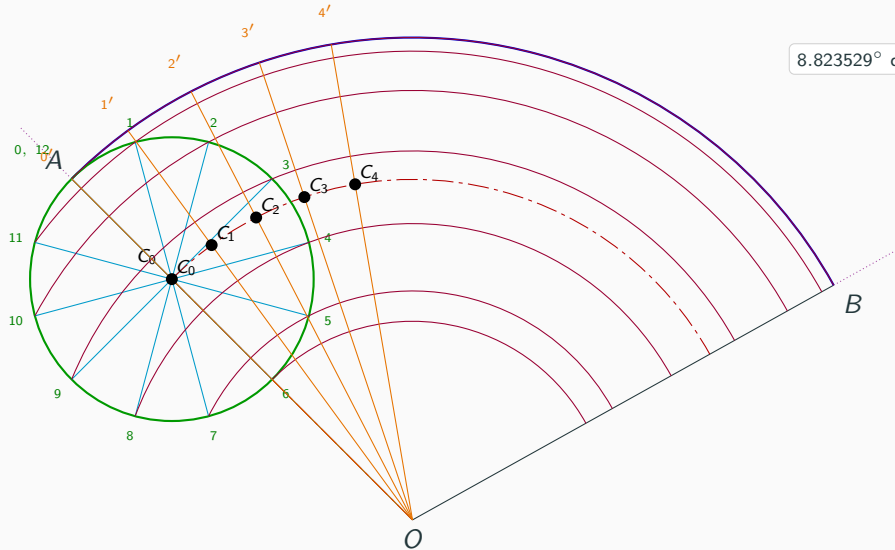
## Step 9: Divide $\angle AOB$ into 12 equal parts



8.823529° divisions.

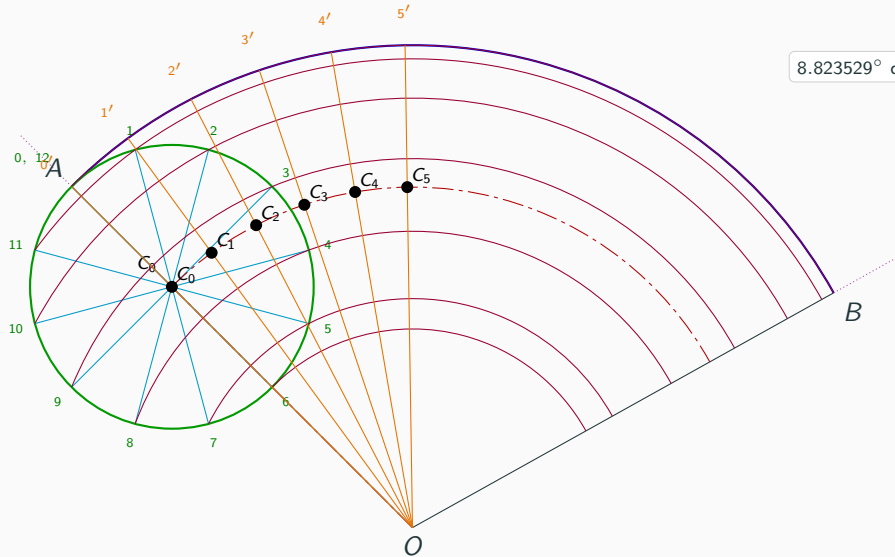


## Step 9: Divide $\angle AOB$ into 12 equal parts

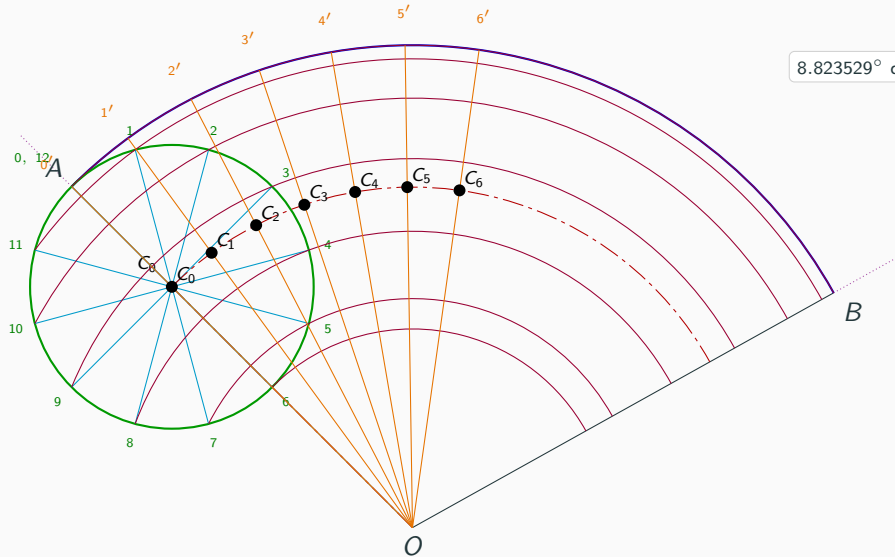


8.823529° divisions.

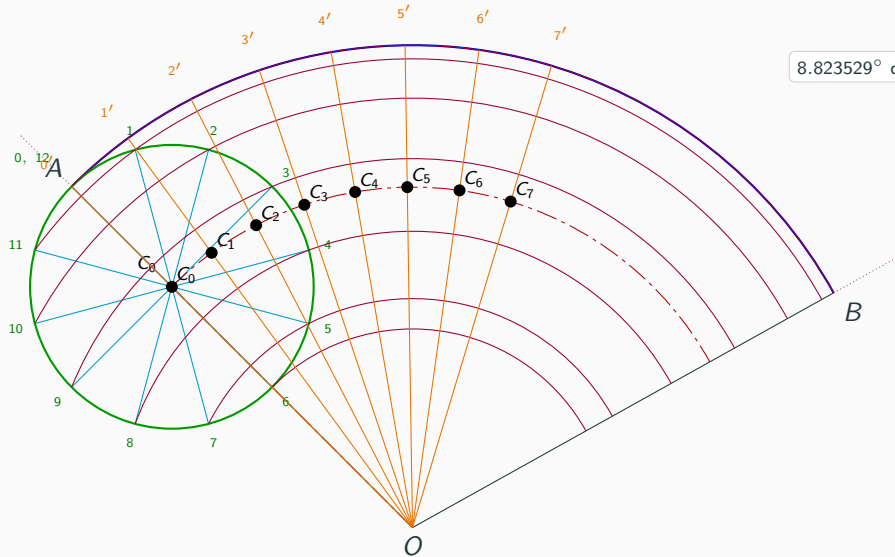
## Step 9: Divide $\angle AOB$ into 12 equal parts



## Step 9: Divide $\angle AOB$ into 12 equal parts

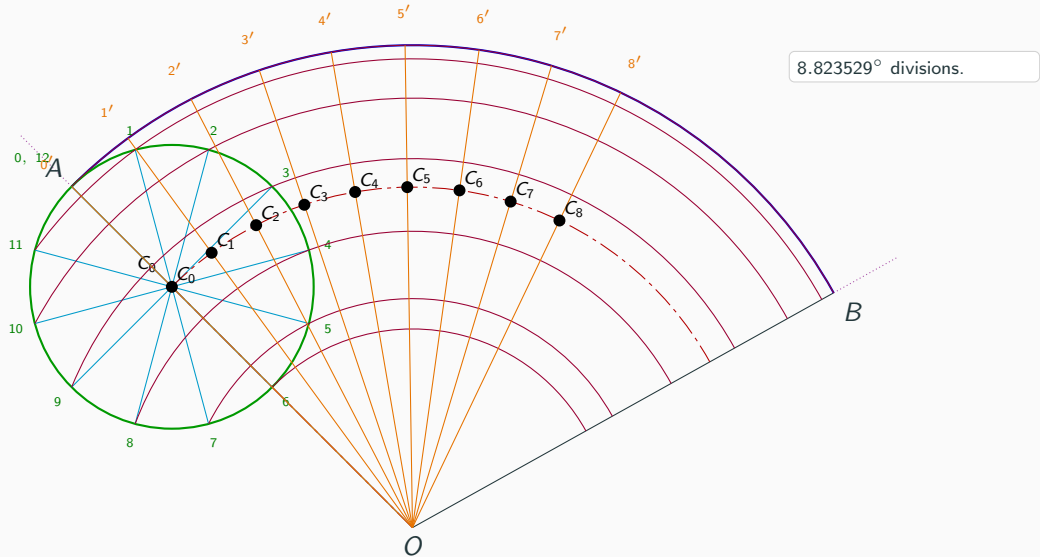


## Step 9: Divide $\angle AOB$ into 12 equal parts

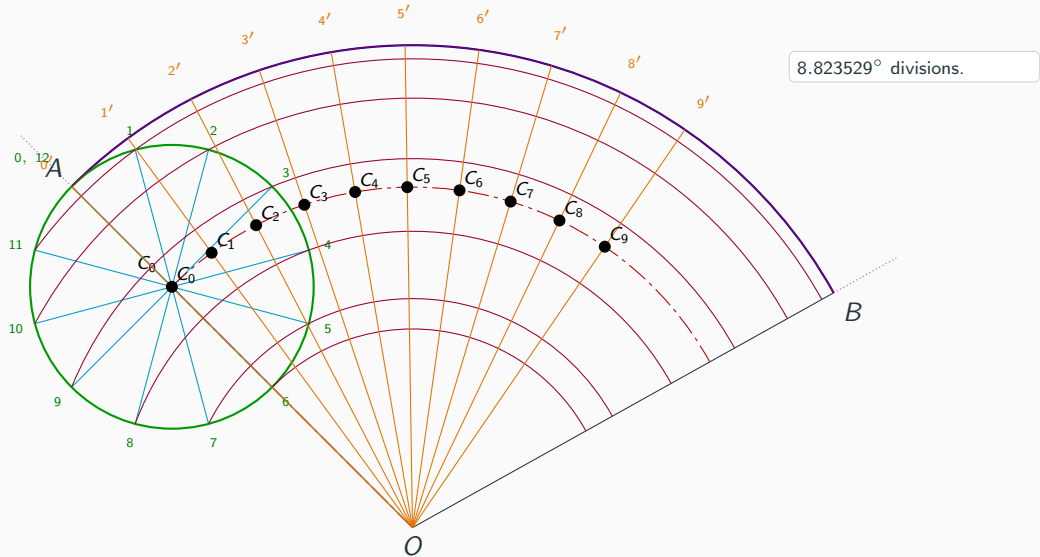


$8.823529^\circ$  divisions.

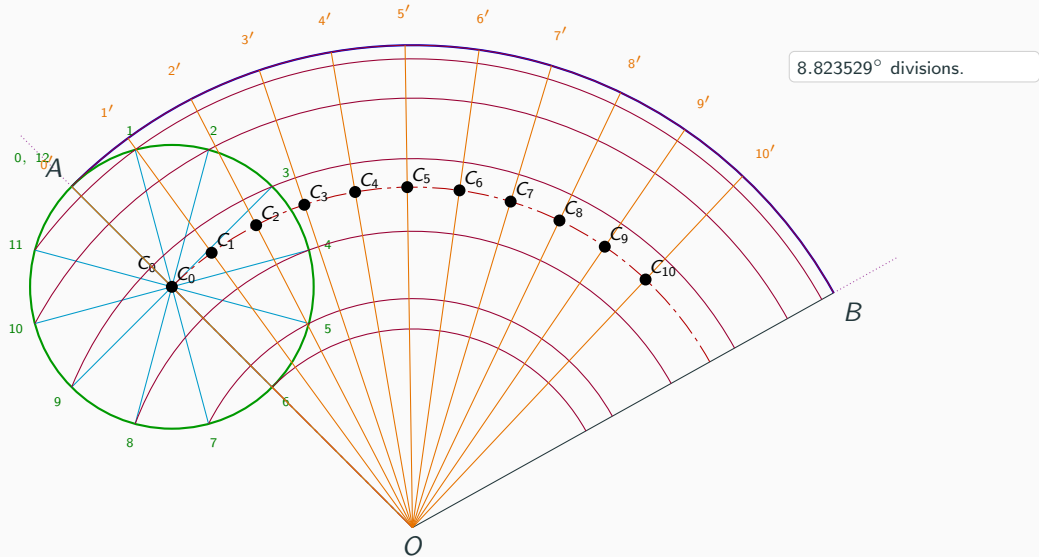
## Step 9: Divide $\angle AOB$ into 12 equal parts



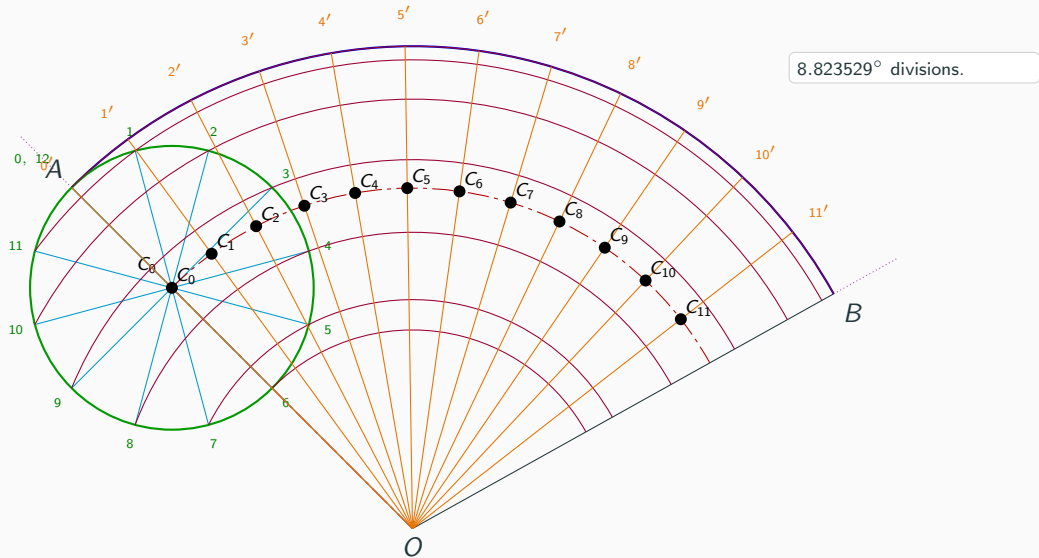
## Step 9: Divide $\angle AOB$ into 12 equal parts



## Step 9: Divide $\angle AOB$ into 12 equal parts

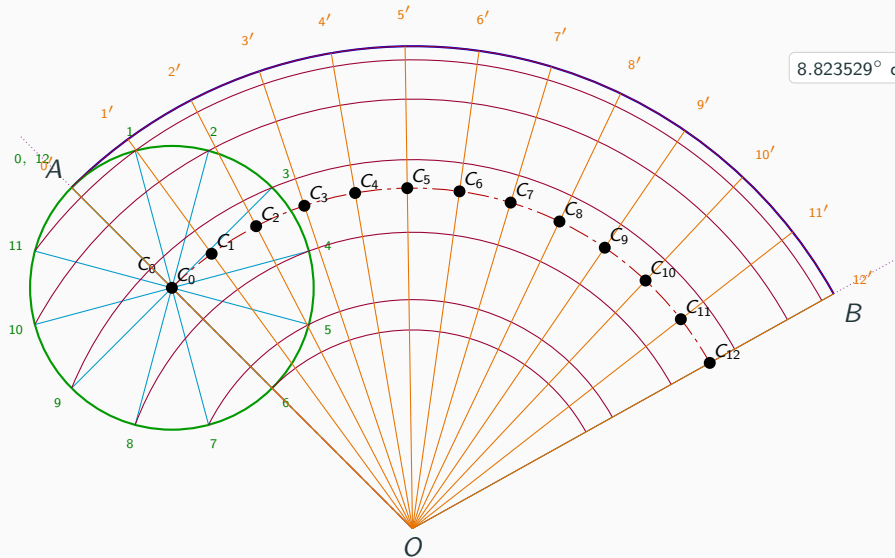


## Step 9: Divide $\angle AOB$ into 12 equal parts

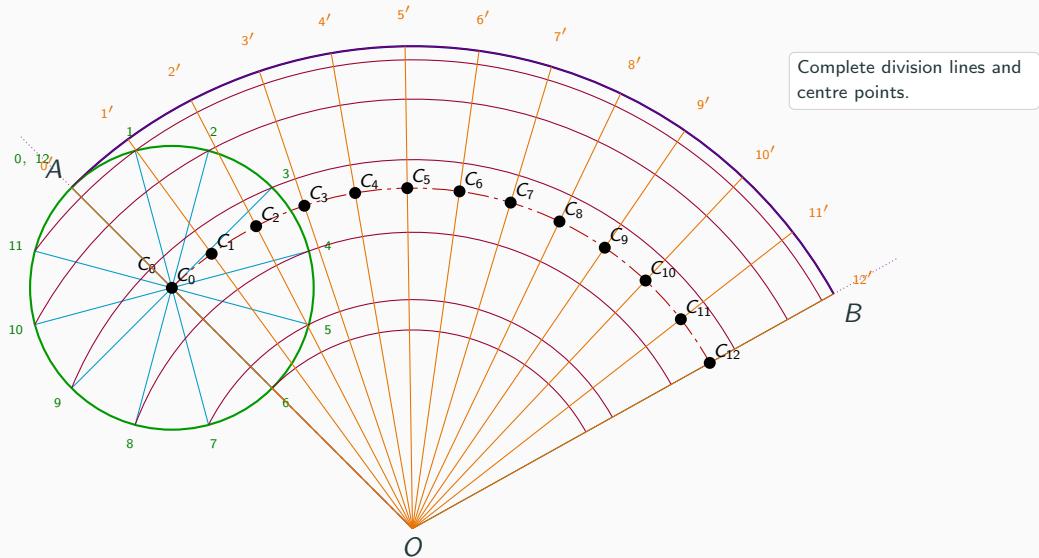




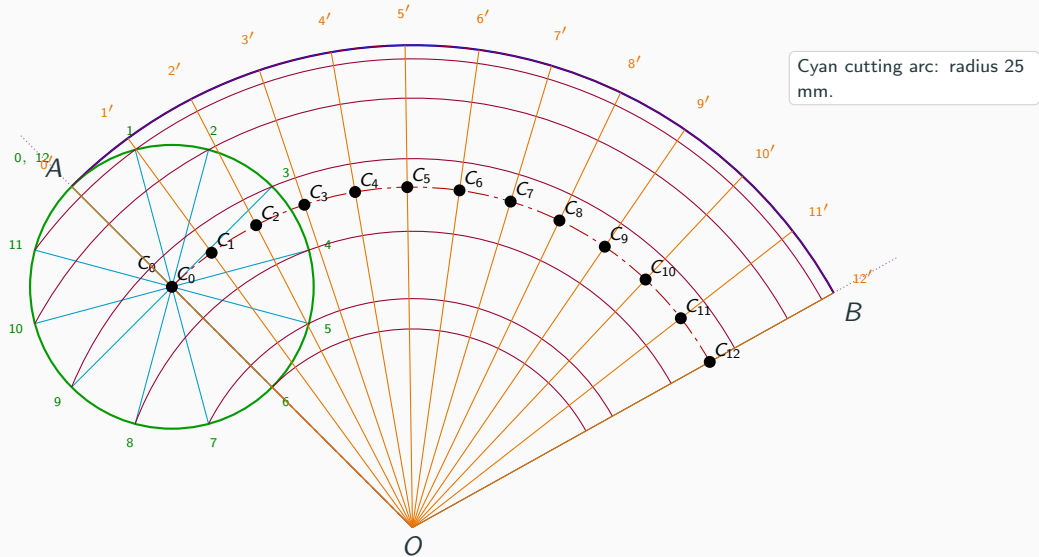
## Step 9: Divide $\angle AOB$ into 12 equal parts



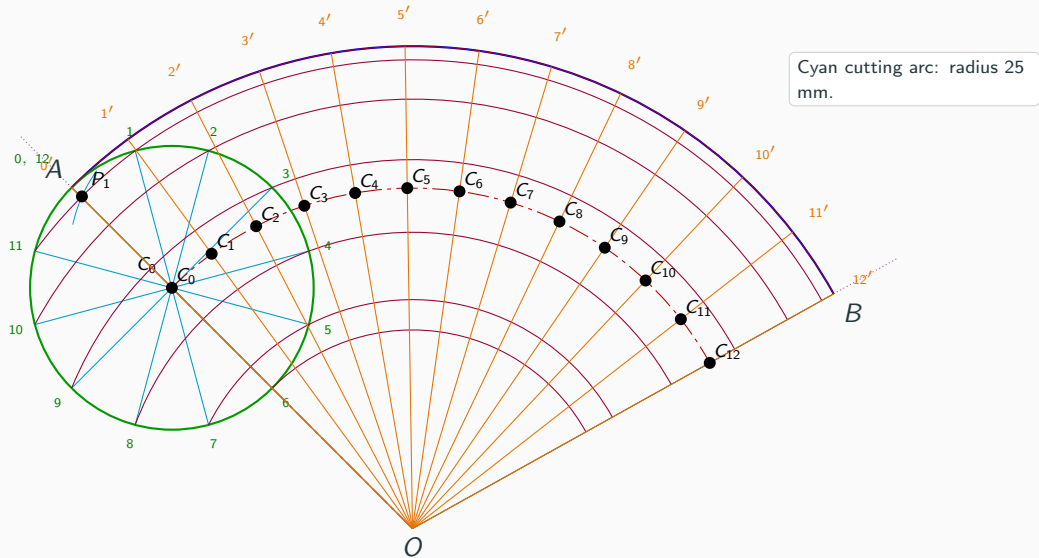
## Step 10: Complete the angular divisions and centre points



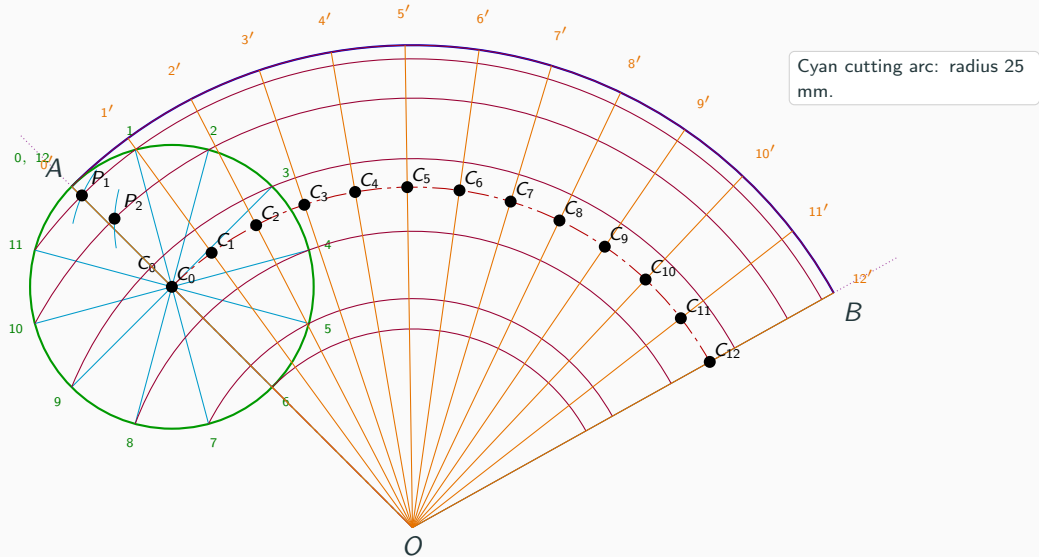
## Step 11: Draw the cutting arc and locate $P_i$



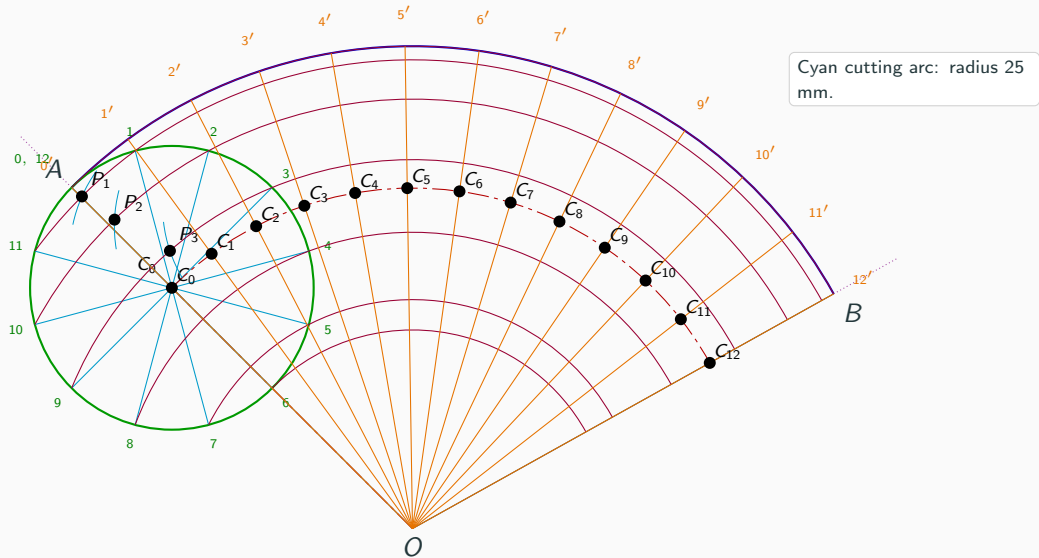
## Step 11: Draw the cutting arc and locate $P_i$



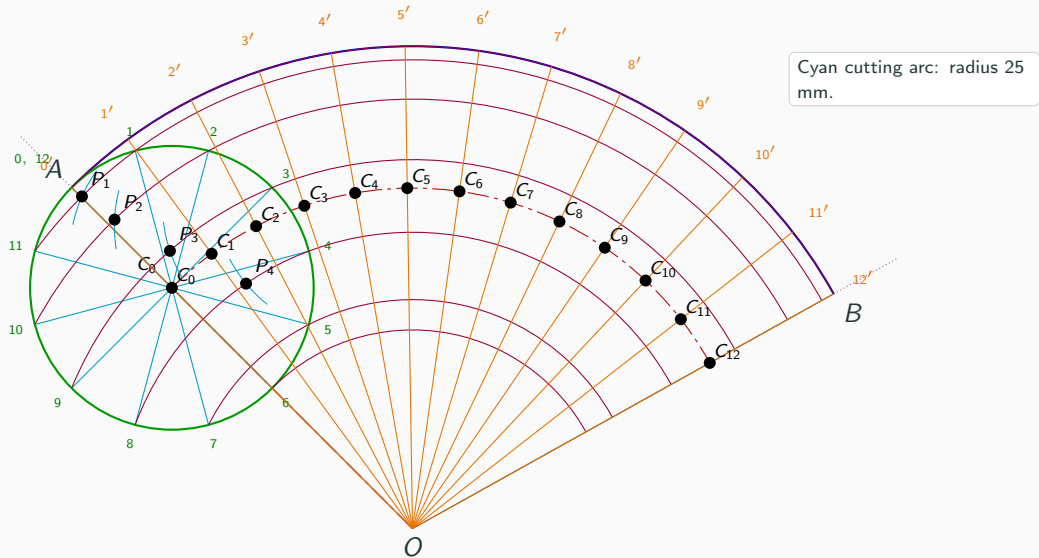
## Step 11: Draw the cutting arc and locate $P_i$



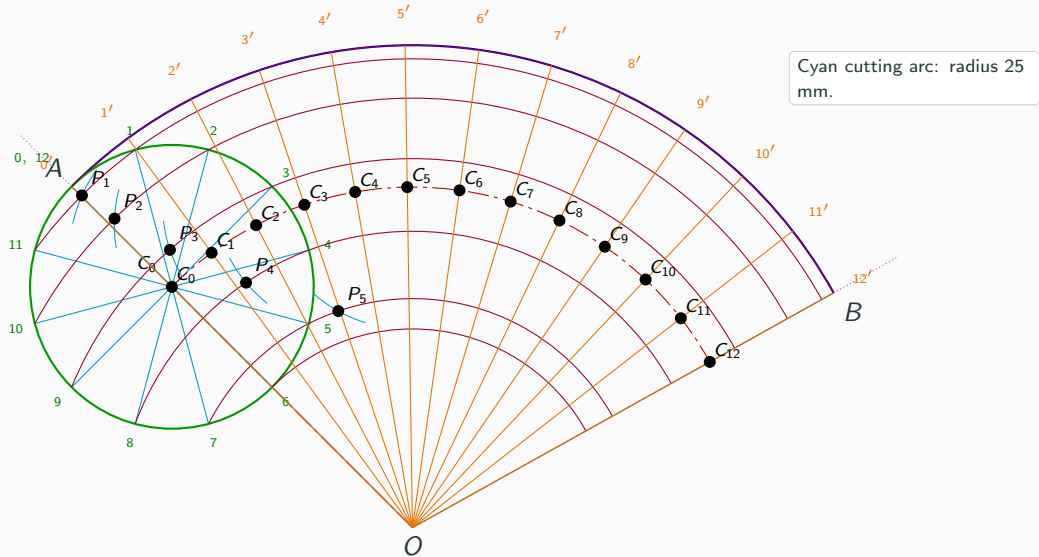
## Step 11: Draw the cutting arc and locate $P_i$



## Step 11: Draw the cutting arc and locate $P_i$

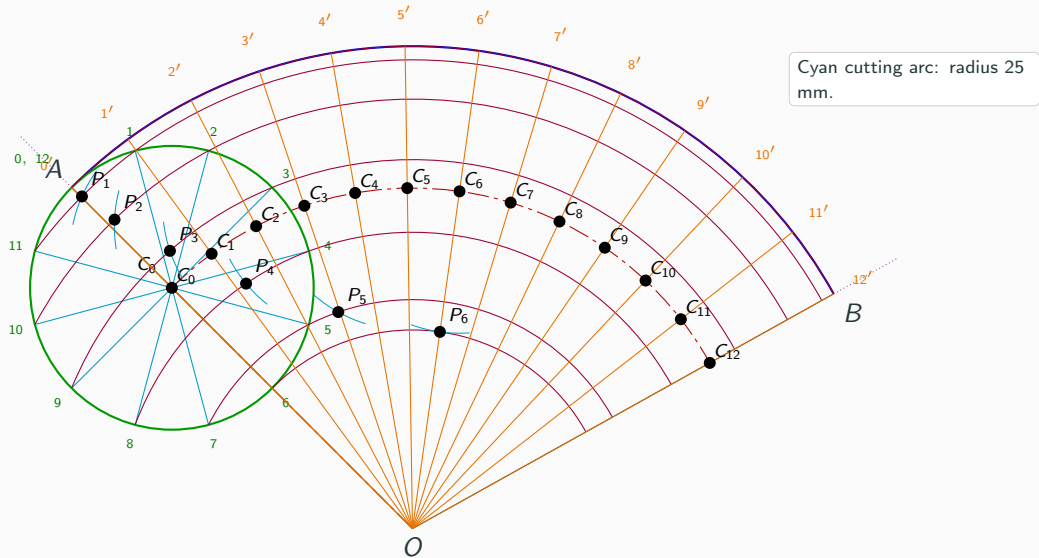


## Step 11: Draw the cutting arc and locate $P_i$

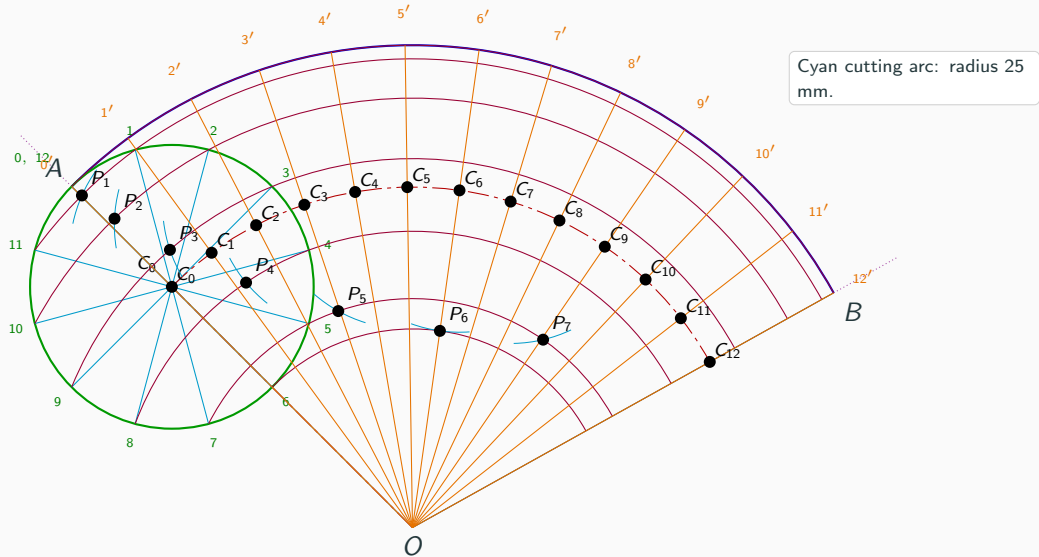




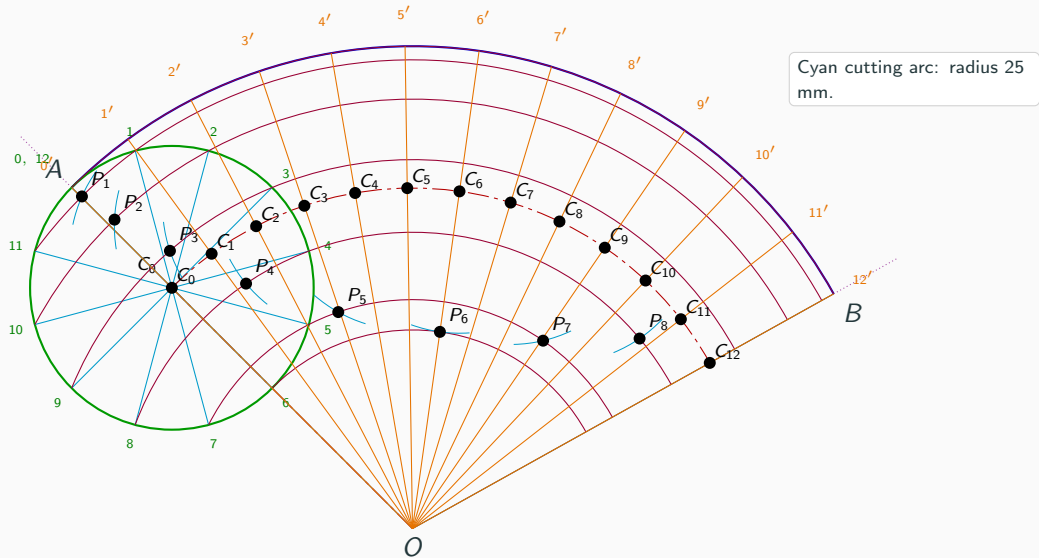
## Step 11: Draw the cutting arc and locate $P_i$



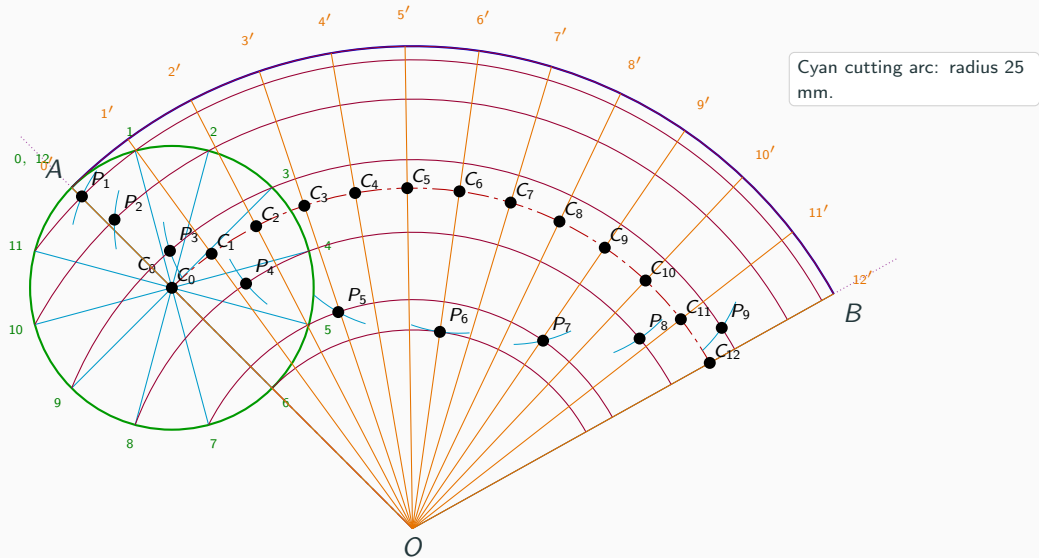
## Step 11: Draw the cutting arc and locate $P_i$



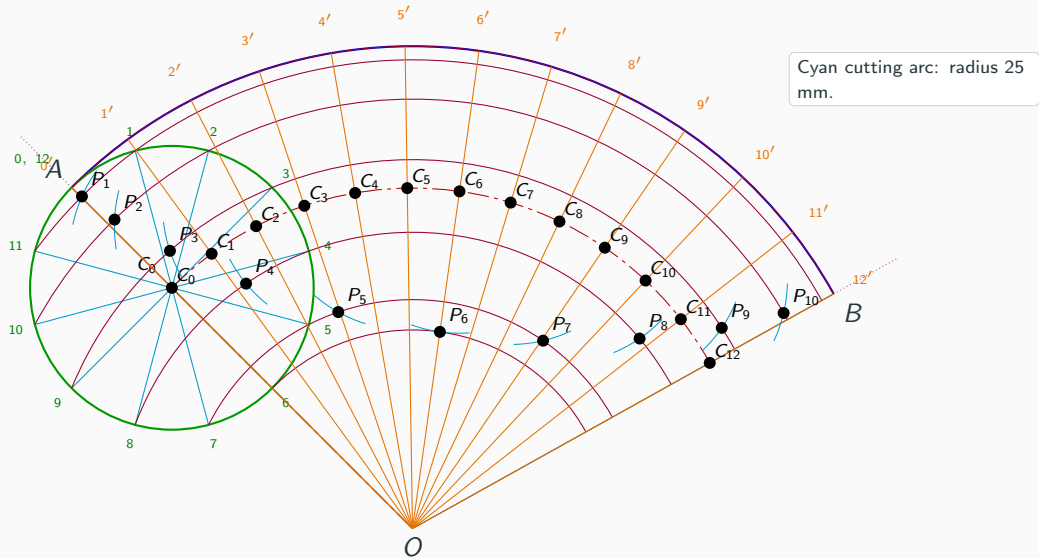
### Step 11: Draw the cutting arc and locate $P_i$



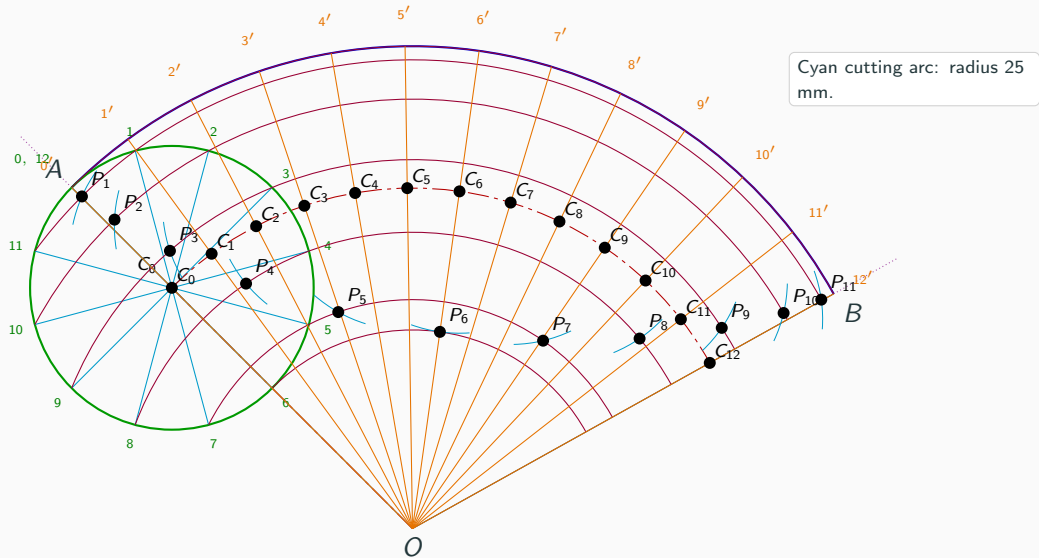
## Step 11: Draw the cutting arc and locate $P_i$



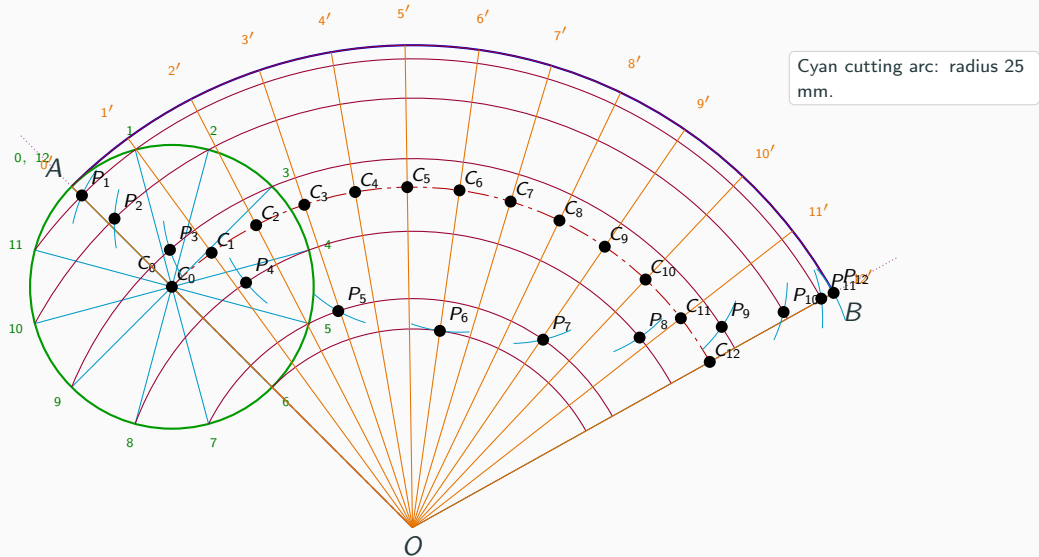
## Step 11: Draw the cutting arc and locate $P_i$



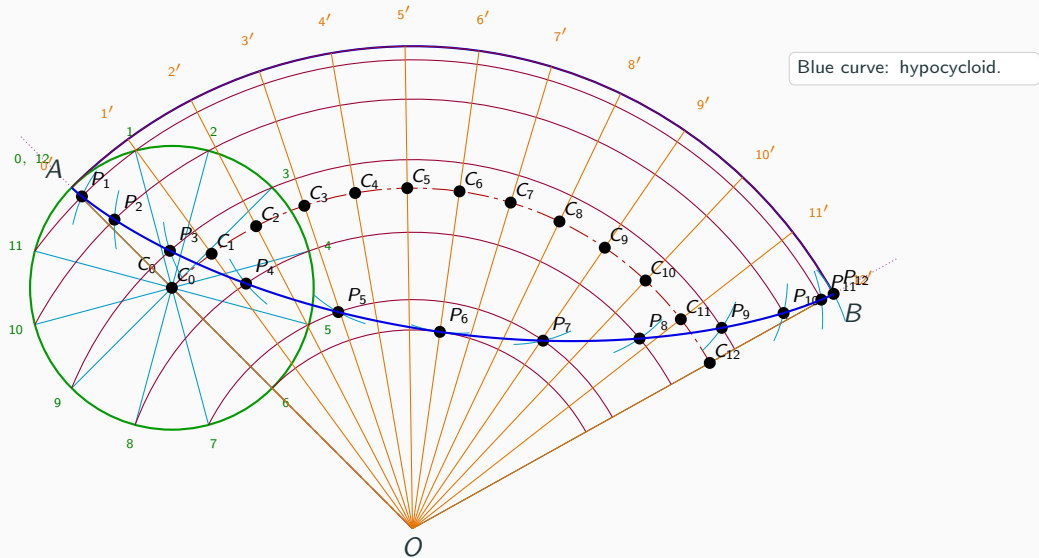
## Step 11: Draw the cutting arc and locate $P_i$



## Step 11: Draw the cutting arc and locate $P_i$

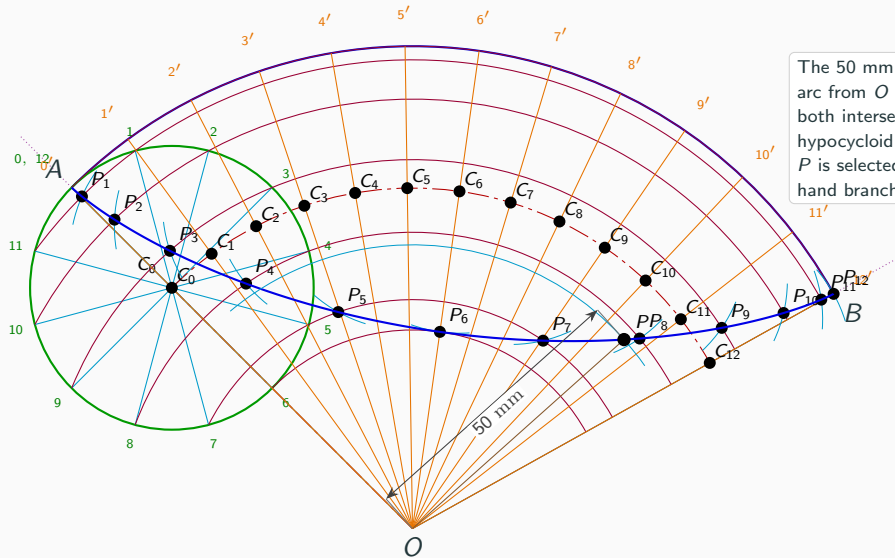


## Step 12: Draw the hypocycloid



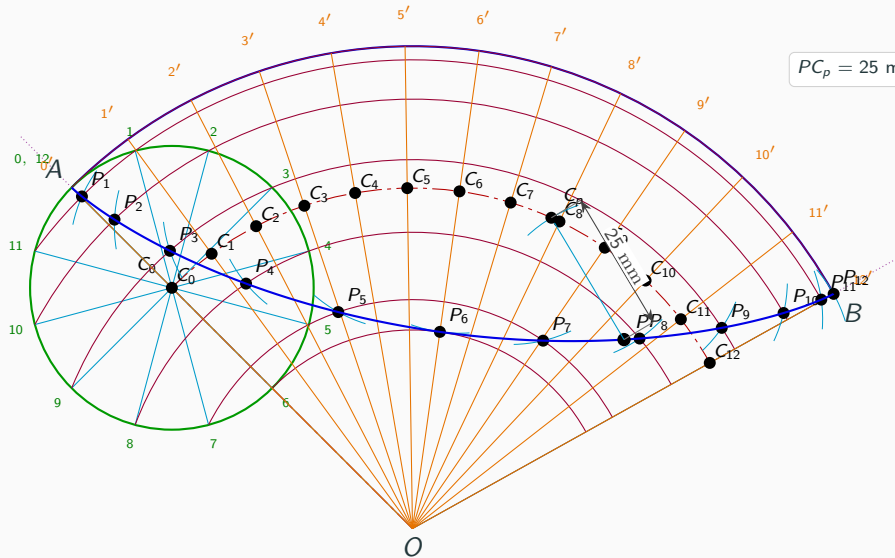


## Step 13: Locate $P$ at 50 mm from $O$

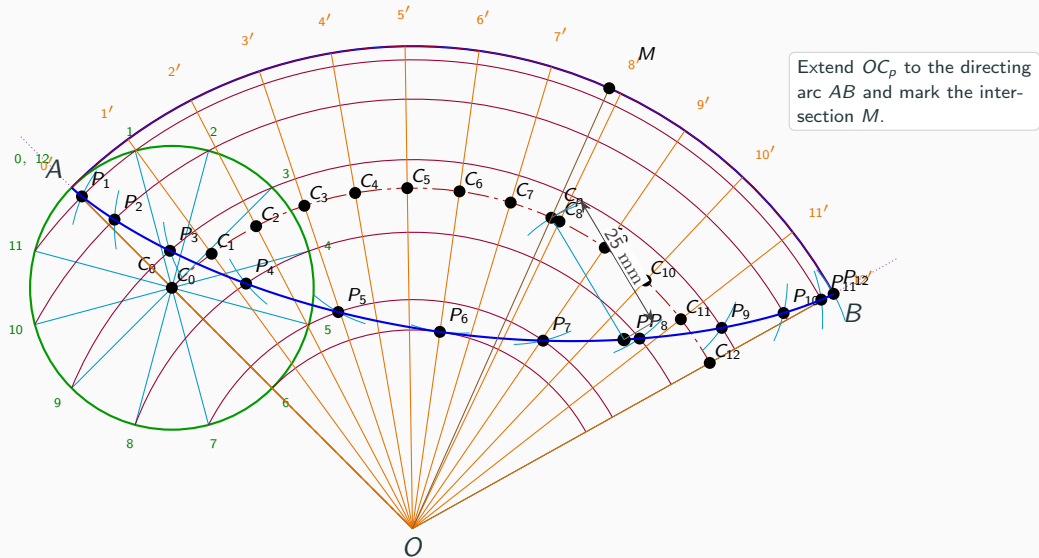


The 50 mm construction arc from  $O$  extends beyond both intersections with the hypocycloid.  $P$  is selected on the right-hand branch.

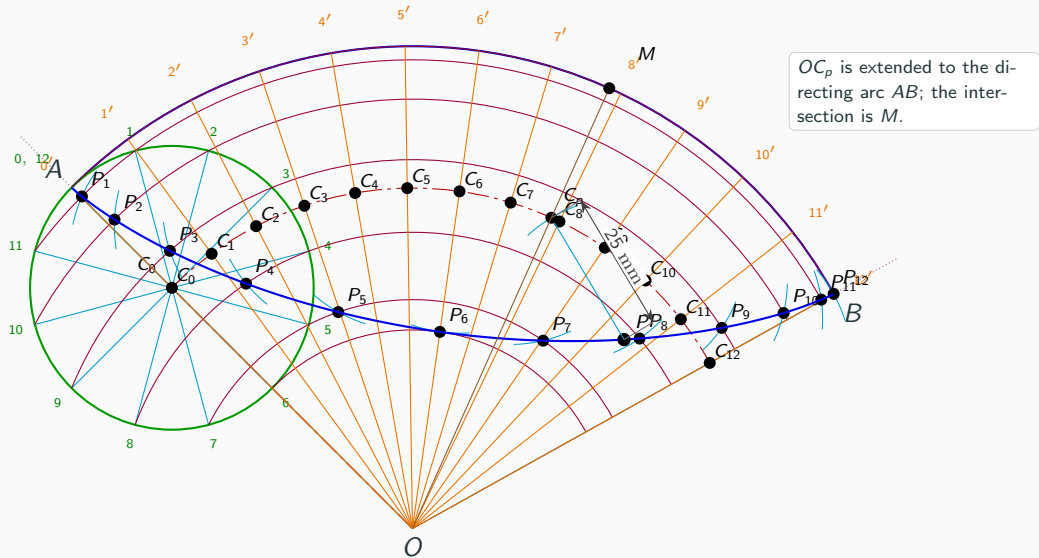
**Step 14: Draw the 25 mm radius to  $C_p$**


$$PC_p = 25 \text{ mm.}$$

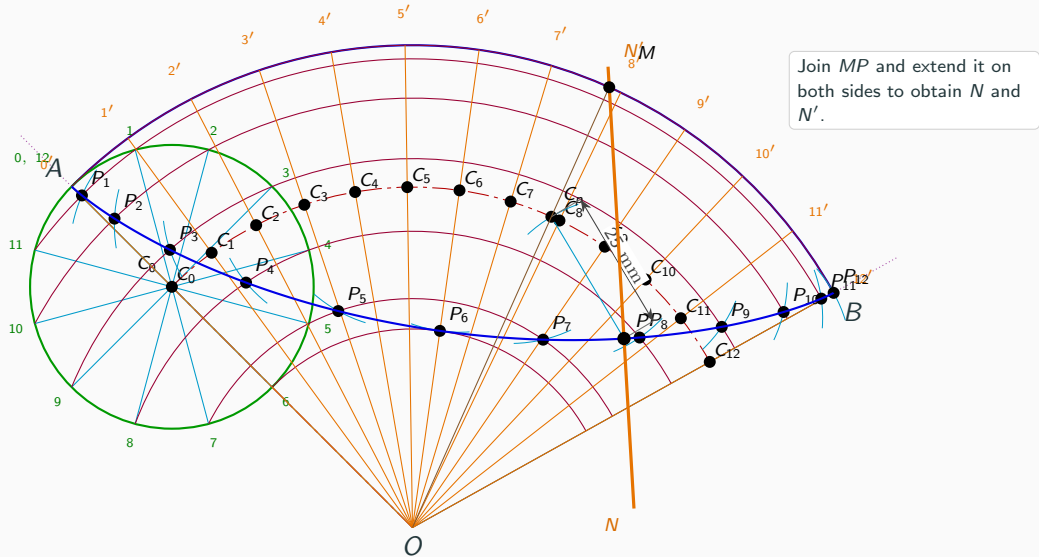
## Step 15: Extend $OC_p$ to the directing arc $AB$ and mark the intersection $M$



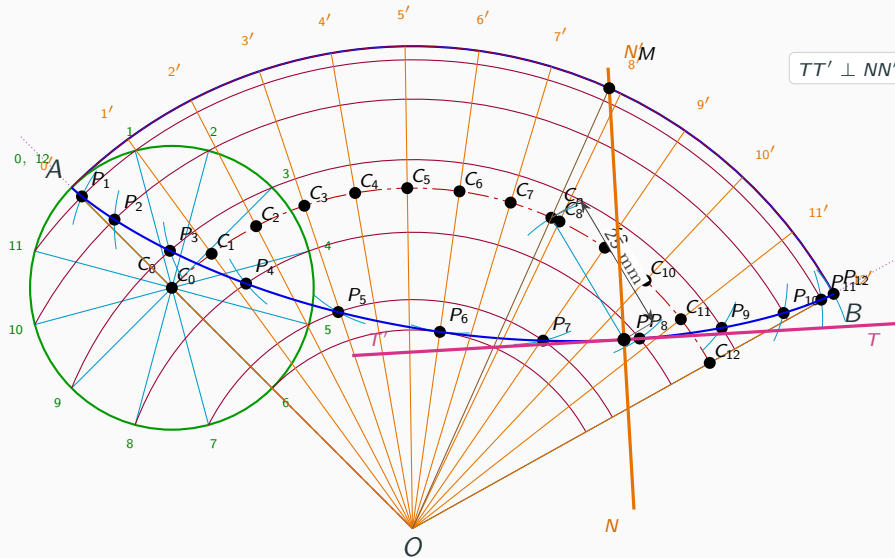
## Step 16: Mark $M$ on the directing circle



## Step 17: Draw the normal $NN'$ through $P$

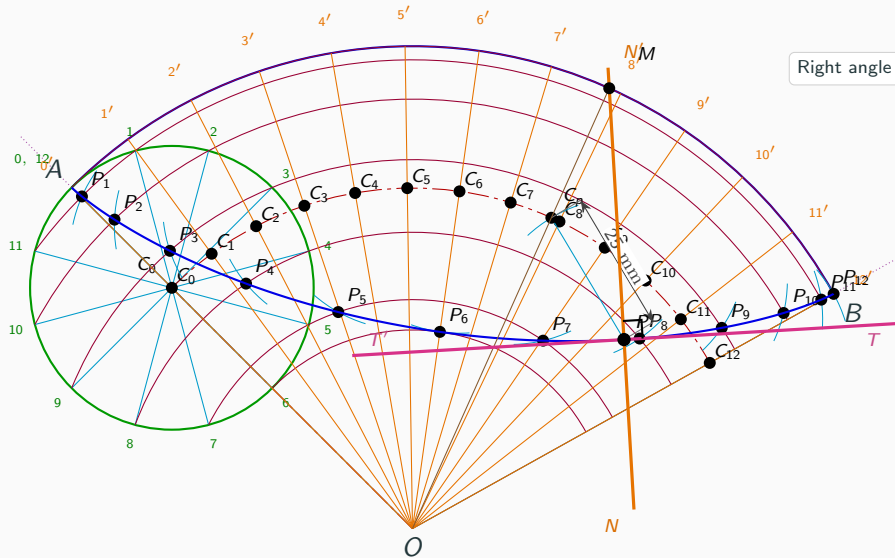


## Step 18: Draw the tangent $TT'$ perpendicular to $NN'$



$TT' \perp NN'$  at  $P$ .

## Step 19: Show the right angle at $P$



Right angle at  $P$ .

## Step 20: Final completed hypocycloid construction

